COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 / Data Structures and Analysis	**SECTION:**	CPE21S4
DATE PERFORMED:	September 18, 2026	**DATE SUBMITTED:**	September 18, 2026
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HANDS-ON ACTIVITY 9.1 [ACTIVITY NAME]

1. SUPPLEMENTARY OUTPUT

MAIN FILE:

main.cpp

```cpp
#include <iostream>
#include "BinSorts.h"

int main(){
    // create the root node:
    BinarySearchTree<int> tree(20);

    tree.insert(10);
    tree.insert(5);
    tree.insert(70);
    tree.insert(80);
    tree.insert(40);
    tree.insert(50);

    std::cout <<  "Pre-Order: " << std::endl;
    tree.preOrder();
    std::cout << std::endl;

    std::cout <<  "In-Order: " << std::endl;
    tree.inOrder();
    std::cout << std::endl;

    std::cout <<  "Post-Order: " << std::endl;
    tree.postOrder();
    std::cout << std::endl;

    return 0;
}
```

BinSorts.h

```cpp
#ifndef BinSorts_h
#define BinSorts_h

template <typename T>
class BinarySearchTree{
    private:
       T key;
       //pointers:
       BinarySearchTree<T>* left;
       BinarySearchTree<T>* right;

    public:
    //constructor:
    BinarySearchTree<T>(T value);
    void insert (T value);


    //triversing order:
    void preOrder ();
    void inOrder();
    void postOrder();

};
```

```cpp
template <typename T>
BinarySearchTree<T>::BinarySearchTree(T value){
    key = value;
    left = nullptr;
    right = nullptr;
}

template <typename T>
void BinarySearchTree<T>::insert(T value){
   if(value < key)
   //do something:
   if (left == nullptr){
       left = new BinarySearchTree<T>(value);
   } else {
       left -> insert(value);
   } else if (value > key ){

    if(right == nullptr){
       right = new BinarySearchTree<T>(value);
    } else{
       right -> insert(value);
    }
   }
}
```

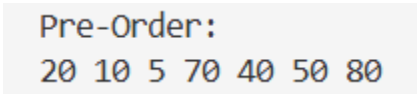
```cpp
template <typename T>
void BinarySearchTree<T>::preOrder(){
 std::cout << key << " " << std::endl;
 if(left != nullptr){
     left -> preOrder();
 }
 if (right != nullptr){
     right -> preOrder();
 }
}


 template <typename T>
void BinarySearchTree<T>::inOrder(){
 std::cout << key << " " << std::endl;
 if(left != nullptr){
     left -> inOrder();
 }
 if (right != nullptr){
     right -> inOrder();
 }
}

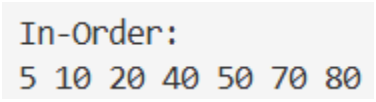
 template <typename T>
void BinarySearchTree<T>::postOrder(){
 std::cout << key << " " << std::endl;
 if(left != nullptr){
     left -> postOrder();
 }
 if (right != nullptr){
     right -> postOrder();
 }
}

#endif
```

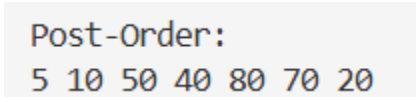
PRE-ORDER

SCREENSHOTS	
OBSERVATION ANALYSIS	- ***Since this is a pre-order tree diagram flow, the root gets printed first and the sequence of the children continues (according to the statements inside the program). The program then visits the left child first, followed by the right child, following the order of the statements written inside the program. This process continues recursively until all the nodes in the tree have been visited.***

IN ORDER

SCREENSHOTS	
OBSERVATION ANALYSIS	- ***This is an in-order sequence where the root is printed in the middle of the output. The program first visits and prints the nodes on the left side of the tree, then prints the root node, and finally continues to the right side of the tree.***

POST ORDER

SCREENSHOTS	
OBSERVATION ANALYSIS	- ***This is the traverse of the array, where the last child is printed first while the root is as the last of the output display. The program starts by traversing the left child, followed by the right child, while the root is printed last in the output.***

- **Overall, to conclude this activity, I realized and pondered that Tree Diagram is very helpful in organizing data. In which it can be utilized and great help when searching, organizing data, expression trees, file systems, and other applications that require hierarchical data structures.**

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

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Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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